MIL-D-28728/4B
3 September 1986
SUPERSEDING
MIL-D-28728/4A
6 December 1971

MILITARY SPECIFICATION SHEET

DIAL, CONTROL, MULTITURN COUNTERS, VERNIER, 1.16 DIAMETER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the dial described herein shall consist of this specification and the latest issue of MIL-D-28728.

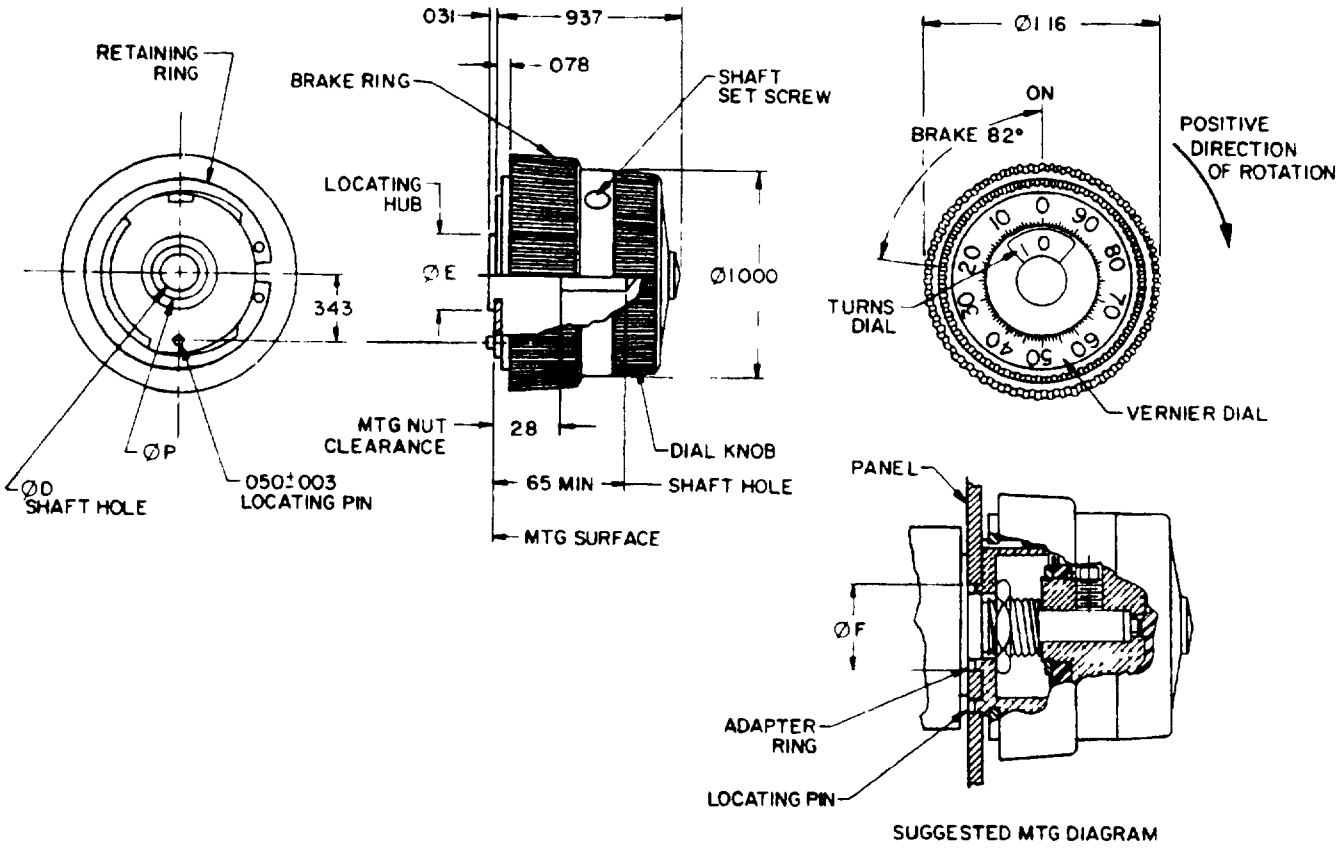


FIGURE 1. DIAL, CONTROL, MULTITURN COUNTERS, VERNIER, 1.16 DIAMETER

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TABLE I. DASH NUMBERS AND DESCRIPTION

DESCRIPTION								
DASH NO	GEAR RATIO CODE XX.X:1 (SEE NOTE 5)	COLOR CODE 1/			ØD +.002 -.000	ØP +.010 - 000	MTG NUT ONE REQUIRED	E + 000 -.005 F +.005 -.000
		BRAKE RING	KNOB	DIALS				
001	10.0	B	C	G	.127	.283	.250-32	.375
002	10.0	B	C	B	.127	.283	.250-32	.375
003	10.0	B	C	G	.252	.408	.375-32	.500
004	10.0	B	C	B	.252	.408	.375-32	.500

1/ B = Black; C = Satin chrome; G = Gray; W = White

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NOTES:

1. All dimensions are in inches.

2. Unless otherwise specified, tolerances are $\pm$.010 for three place decimals, $\pm$.02 for two place decimals and $\pm 5^{\circ}$-0' on angles.

3. Numerals: 0.08 high and graduations are indented and color filled. White on black dial and black on gray dial.

4. Indicia:
   Vernier scale - graduations every 18° with subdivisions every 3.6°. Numerals from 0 to 90 in tens, marked every other graduation.
   Turns scale - numerals with optional graduations to indicate number of turns. Two numerals must be visible in the window at all times.

5. Gear ratio: The first two digits of the gear ratio code indicate whole number of turns of the knob needed to rotate the shaft one full turn (see Table I).

REQUIREMENTS:

1. Material: All components shall be of suitable materials to meet requirements.
   Window lens - clear plastic.

2. Finish (see Table I):
   Color in accordance with the following:
   Chrome satin, natural in accordance with QQ-C-320, Class 1, Type II.
   Black Lusterless Paint, color number 37078 in accordance with FED-STD-595.
   Gray Lusterless Paint, color number 36231 in accordance with FED-STD-595.
   White Lusterless Paint, color number 37875 in accordance with FED-STD-595.

3. Operating temperature range: -55° C to 85° C.

4. Weight: 8 ounces maximum.

5. Torque: Operational - 2.0 inch ounce $\pm$15%.
   Locking brake - 25 inch ounce minimum.

6. Method of count: Continuous rotation of vernier scale clockwise for increase 0 to 99 to 0 in count and counter-clockwise for decrease in count. One complete turn of knob and vernier scale shall move turns dial one increment.

7. Readout accuracy: .01 percent of full count (1 part in 1000).

8. Life: 1,000,000 cycles.

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9. Operational speed: 1500 RPM maximum.

10. Part numbers: The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table I.

Example: M28728/4-003 indicates - Dial, ten turn counter, vernier 1.16 diameter, black brake ring, satin chrome knob and gray dials, with .250 diameter shaft hole.

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Navy - AS, OS, SH
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(Project 5355-0192)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-D-28728/4B Rev B	DIAL, CONTROL, MULTI TURN COUNTERS

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street City State ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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